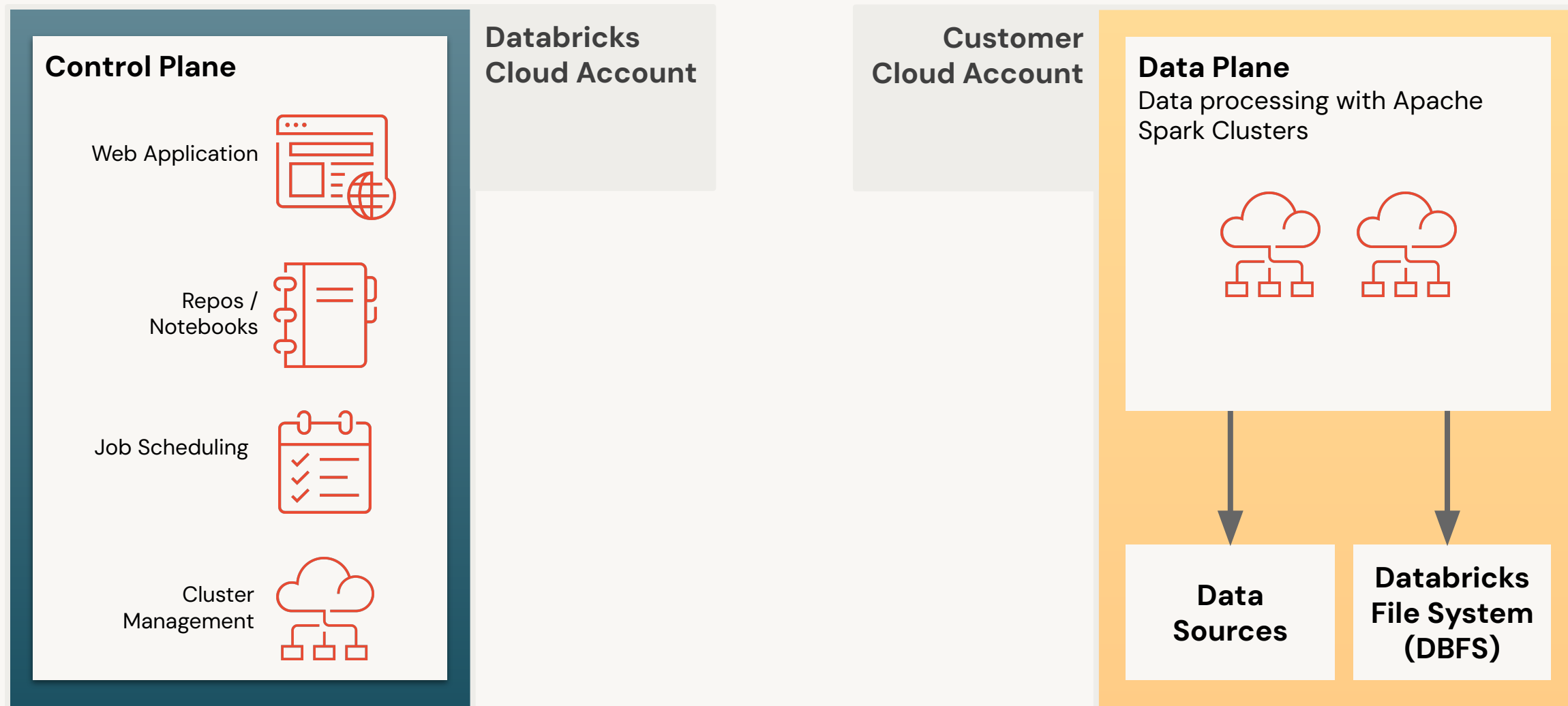


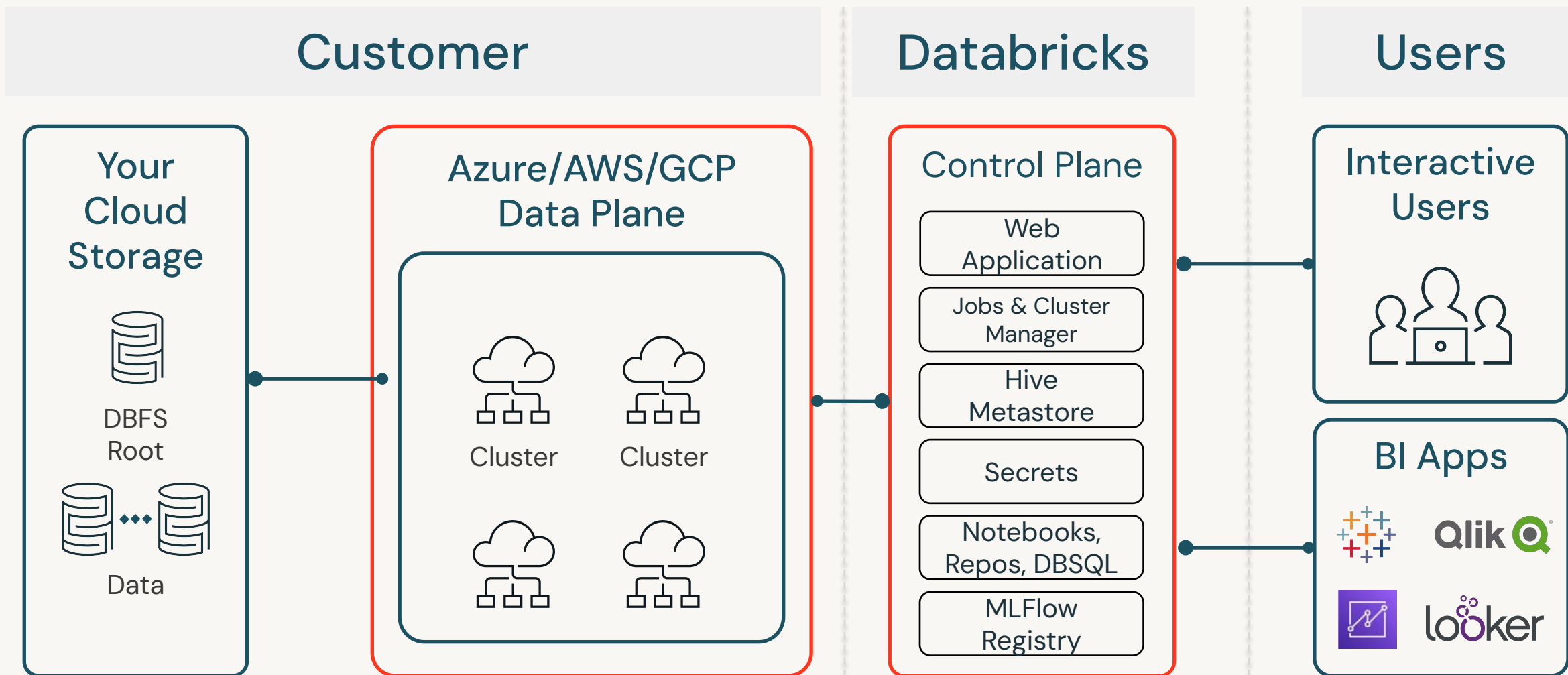


Databricks Architecture & FinOps Cost Management

Clusters

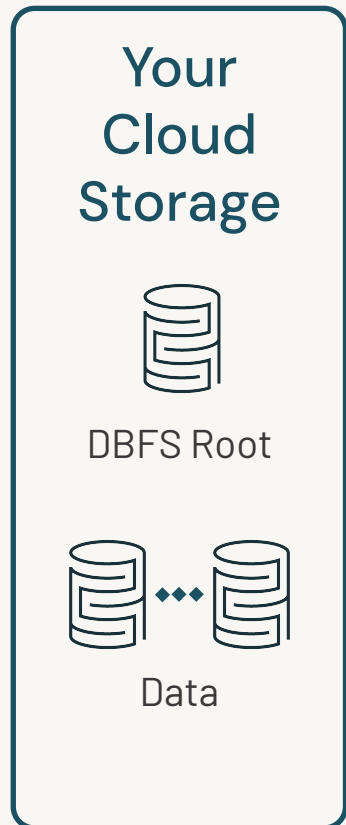


Standard Cluster Architecture

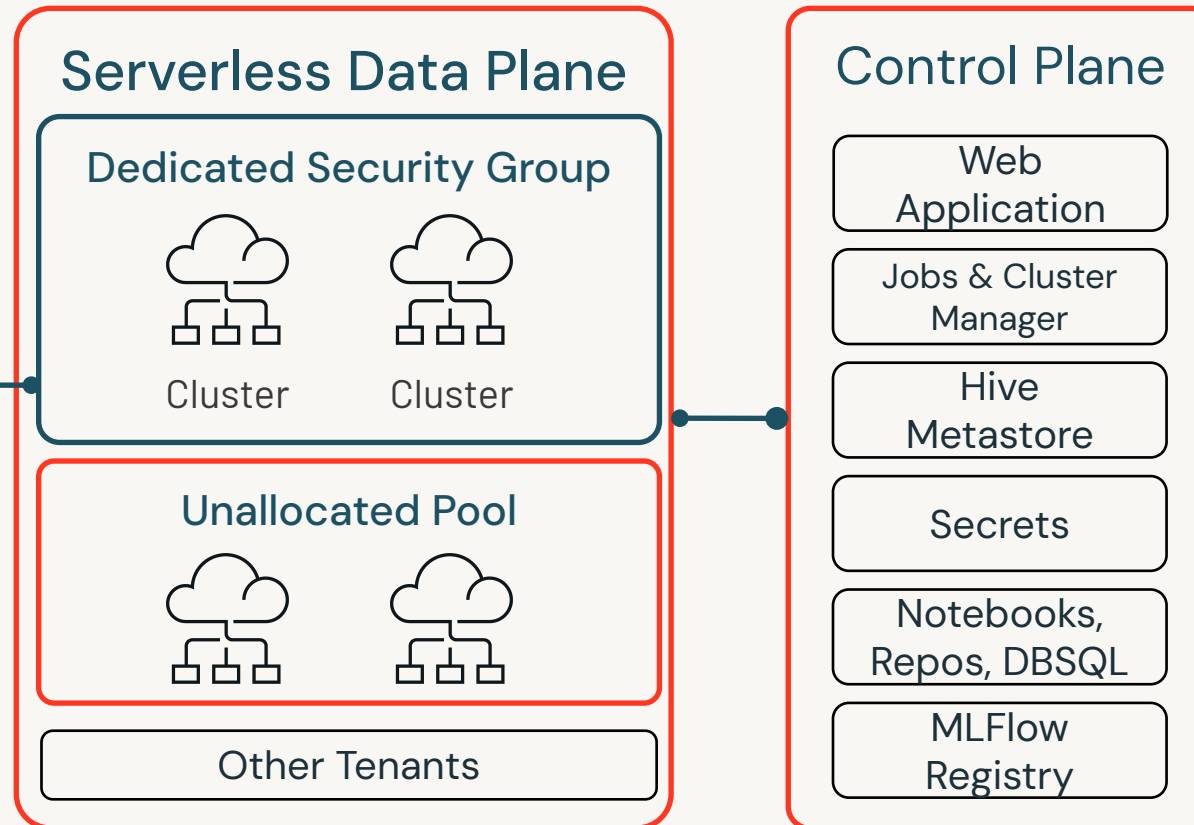


Serverless Architecture

Customer



Databricks



Users

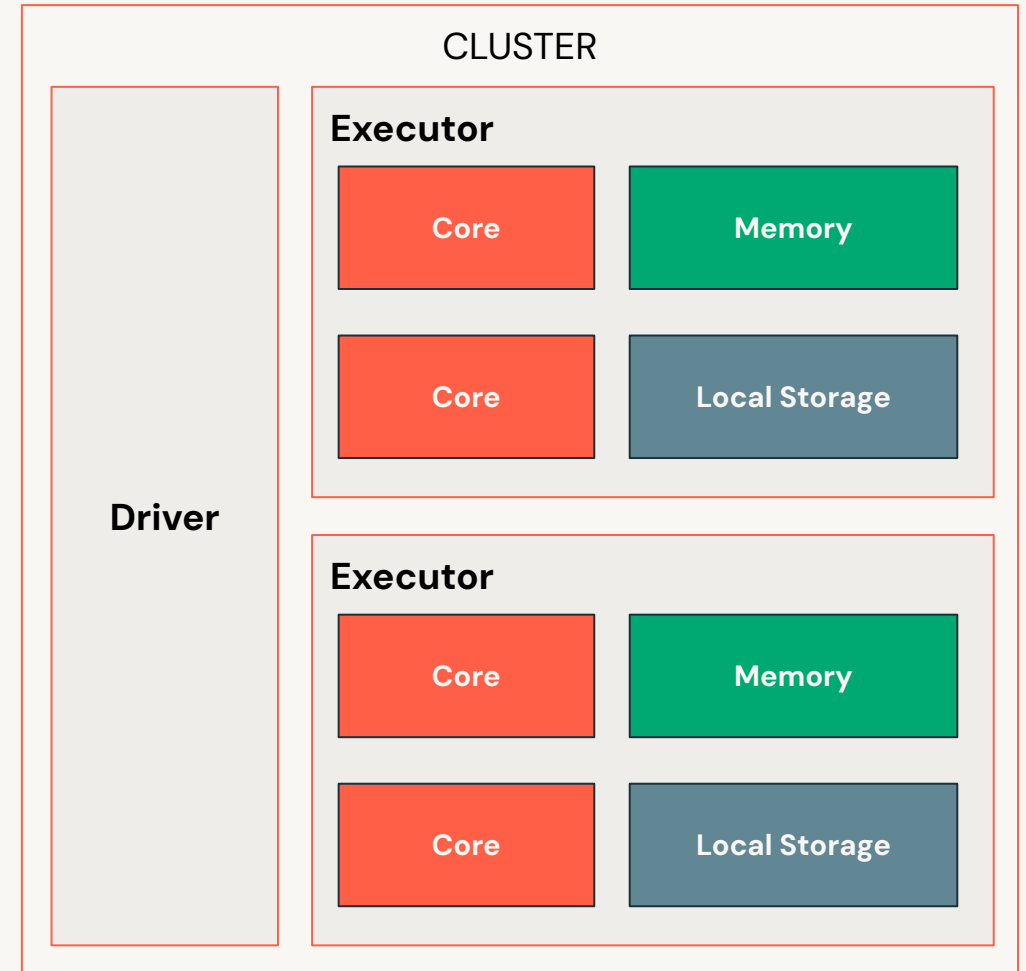


Clusters

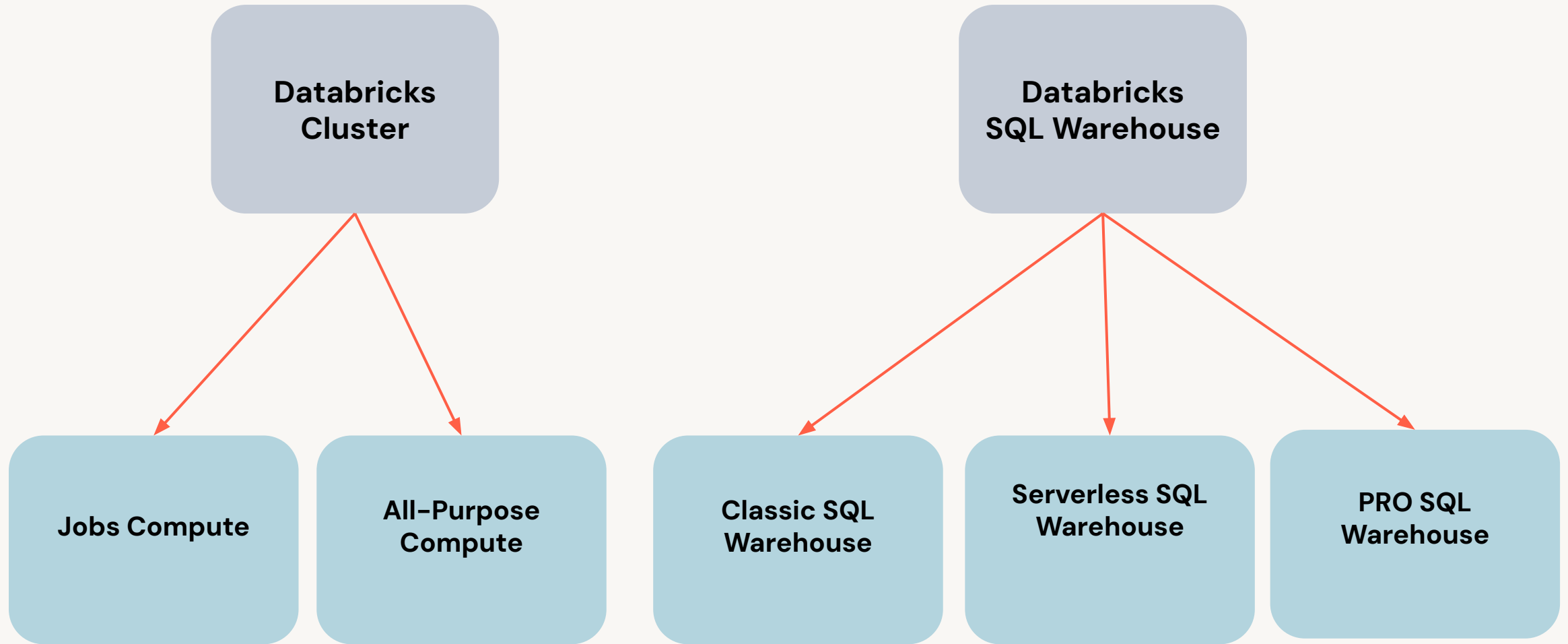
Clusters are made up of one or more virtual machine (VM) instances over which loads are distributed

Driver coordinates activities of executors

Worker (Executors) run tasks composing a Spark job



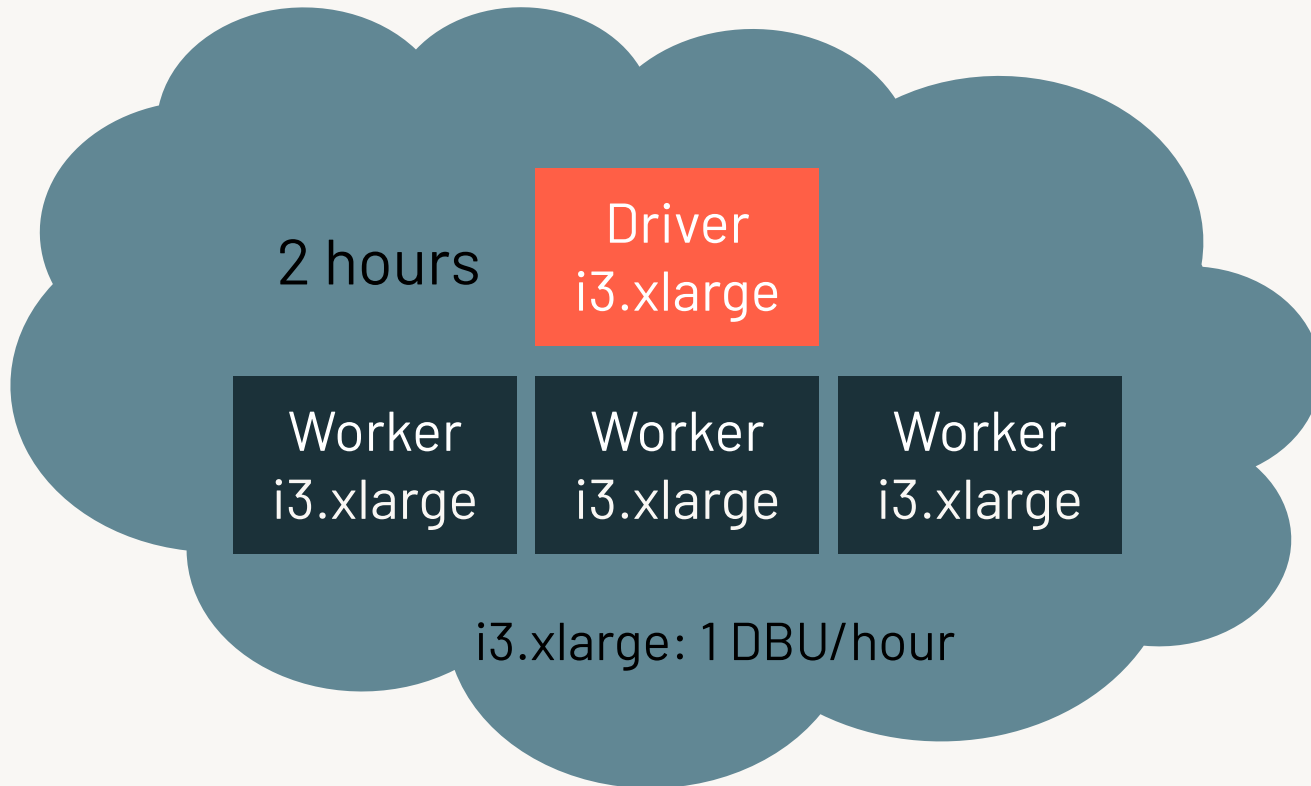
Compute



What is a DBU?

Databricks Unit (DBU)

- A unit of processing power per hour
 - Billed on per-second usage



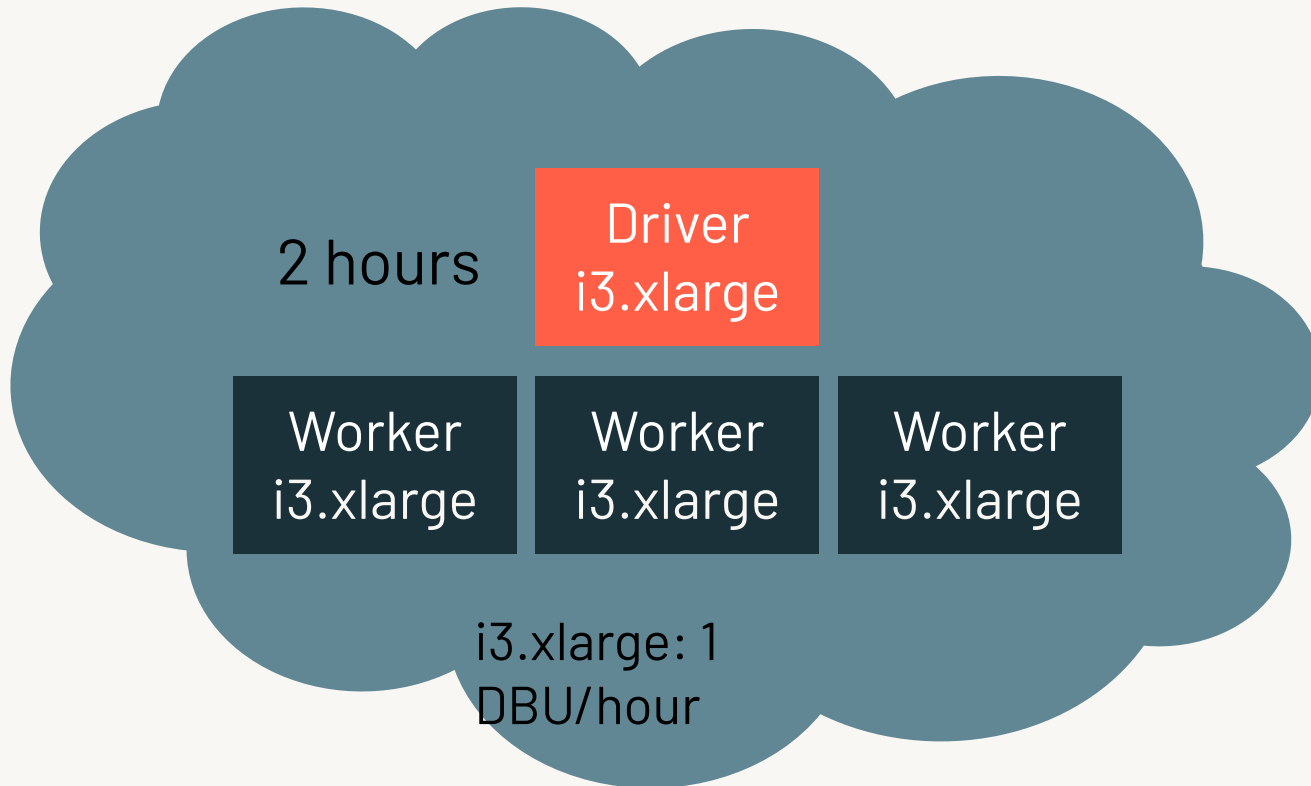
Databricks cost:

□ DBUs?



Databricks Unit (DBU)

- A unit of processing power per hour
 - Billed on per-second usage



Databricks cost:
 $4 \text{ nodes} \times 2 \text{ hours} \times 1 \text{ DBU/hour} = 8 \text{ DBUs}$



Using Databricks Compute Efficiently

ALL PURPOSE COMPUTE

- Also referred to as Interactive
- Data Analysis and ad-hoc exploration
- Shared clusters but best practice is to separate by team or workload
- Anytime an already-running cluster is utilized (including API or scheduled)
- More expensive

JOBS COMPUTE

- Analysis and ad-hoc work
- Run on ephemeral clusters that are created for the job, and terminate on completion
- Pre-scheduled or submitted via API
- Single-user
- Great for isolation and debugging
- Production and repeat workloads
- Lower cost

SQL COMPUTE

- SQL Queries, Analytics, BI reporting
- Single or multi-users
- Create and Share Dashboards
- Native Connectors with BI tools
- Lowest cost

To Summarize...

Type	What	Why	Cost
All-purpose (Interactive or Ad-hoc or Analytics)	Existing clusters	Business analytics Multi-user (ad-hoc workloads)	\$\$\$
Jobs (Automated or Data Engineering)	New scheduled/API- driven cluster with Databricks Runtime Ephemeral	Single-user Repeatable Sensitive to resource constraints Production	\$
SQL Compute	Existing endpoints	SQL Queries Dashboards and BI reporting Connectors with BI tools	\$\$





FinOps

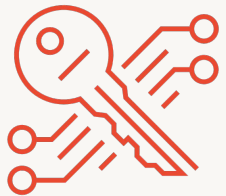
Cost management & value tracking

March 2026

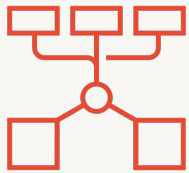




Steps to cost management success



Access



Organize

Setup



Report

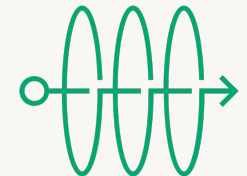


Forecast

Monitor



Optimize



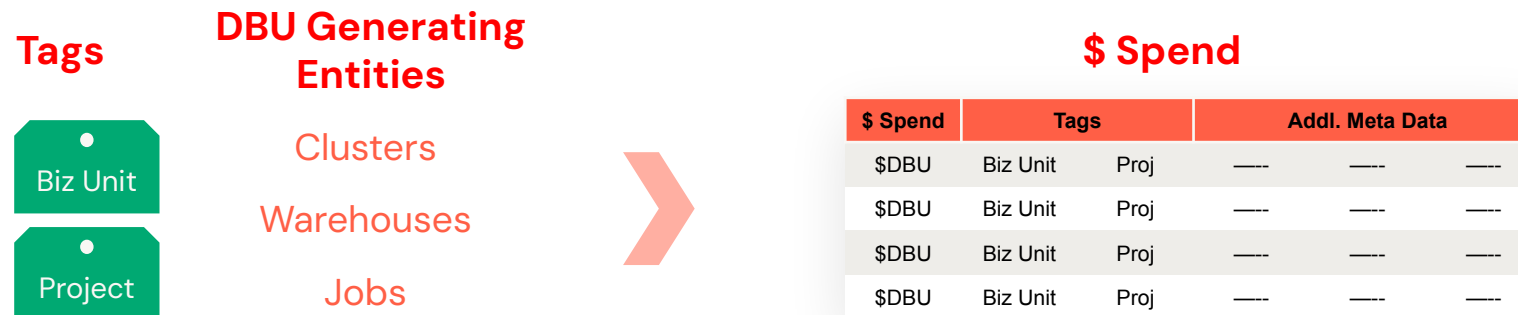
Control

Manage

Cost management – Overview

via Databricks' FinOps best practices

Step 1: Add tags to DBU generating entities & see tags against usage



Step 2: Monitor costs with budgets, review usage in Cost Dashboard, & get quick answers about spend from Genie



Manage costs
with budgets



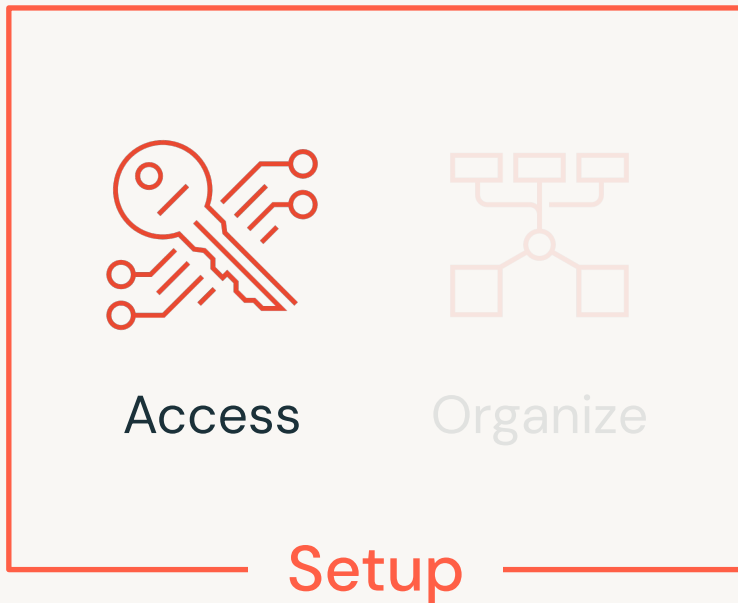
Inspect costs via
cost dashboard



Quick insights on
spend with Genie



Implementation requirements



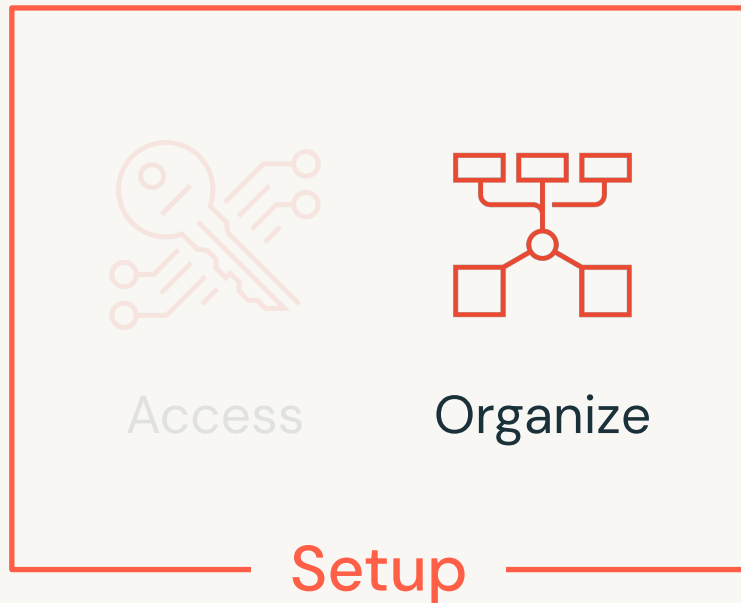
Leadership determines practices and policies

A **Platform Admin** will need to complete enforcement tasks which require appropriate permissions.

Note: Unity Catalog must be setup in one or more workspaces to have an active metastore.



Jobs-to-be-done for cost management



Organizing is the critical step

Here are the core components

1. Determine which tags and values will be required
2. Determine budget amounts for tag keys
3. Admin
 - a. Implements policies to enforce tagging
 - b. Configures Budgets via account console
4. Communications released to Databricks users

Step 1: Determine which tag keys and values will be required



What are tags?

Key: Value pairings

Key	Business Unit
Value	101 (finance)
	102 (legal)
	103 (product)
	104 (sales)
	105 (internal engineering)
	106 (product engineering)

Key	Team
Value	Violet
	Indigo
	Crimson
	Puce
	SI-BoozAllenHamilton
	SI-Credera

Key	Project
Value	Armadillo
	BlueBird
	Rhino
	Dolphin
	Lion
	Eagle

Tagging best practices

Required, optional, and default

Required or General Keys:

What are must have keys for tracking cost? “How much did we spend on X?”

Recommended: Project, and BU (or Team)

Optional:

For more granular insights, apply select high specificity keys (e.g. Version, Builder, Product, Service, Customer, Other / Custom)

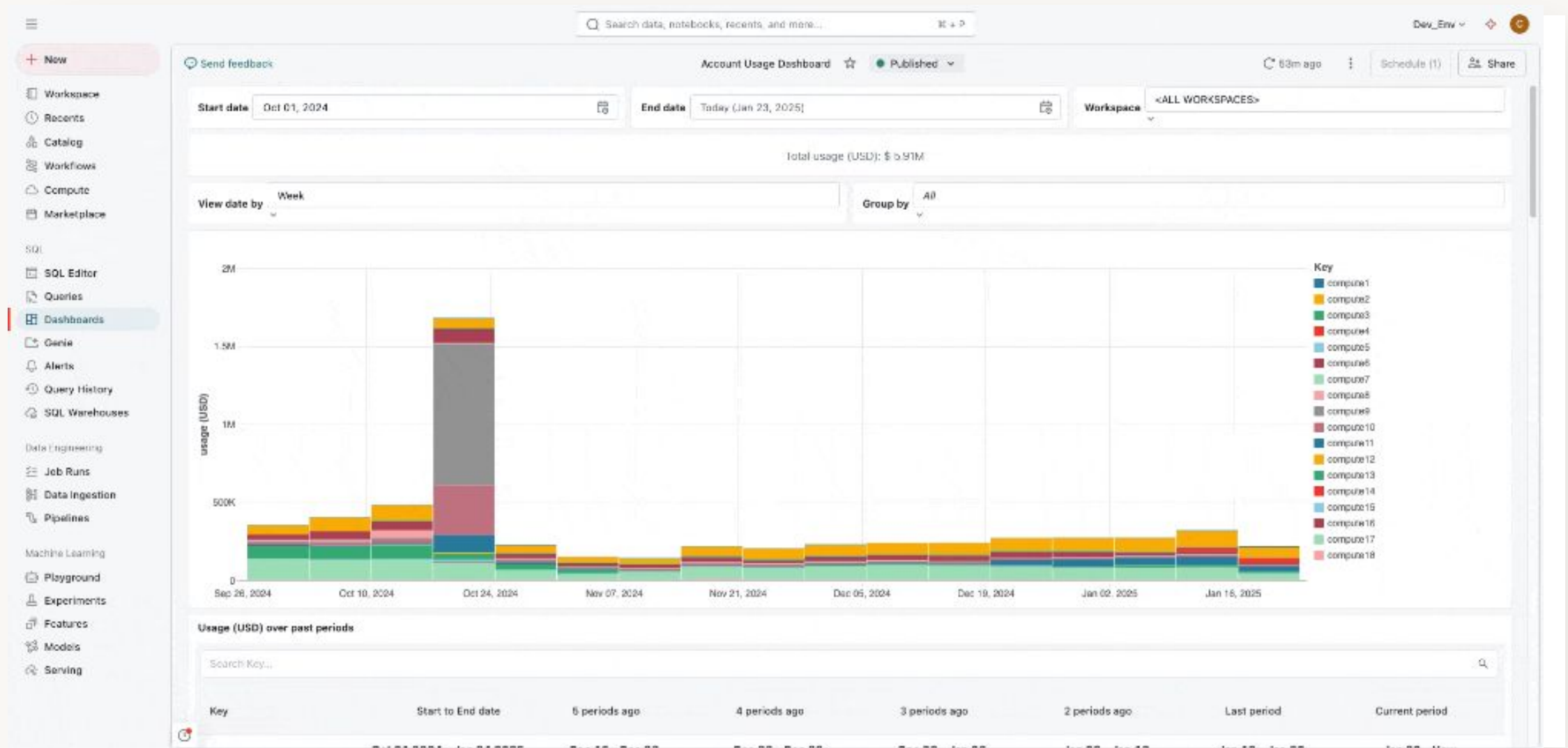
Common: Environment (dev/test/prod), Product, Milestone

Default:

Databricks adds additional tags to resources by default ([list here](#))

Examples:

Tag key	Value
Vendor	Constant value: Databricks
ClusterId	Databricks internal ID of the cluster
ClusterName	Name of the cluster
Creator	Username (email address) of the user who created the cluster



Step 2: Determine budget amounts for tags





Budget Amounts

How much should each *key* be using?

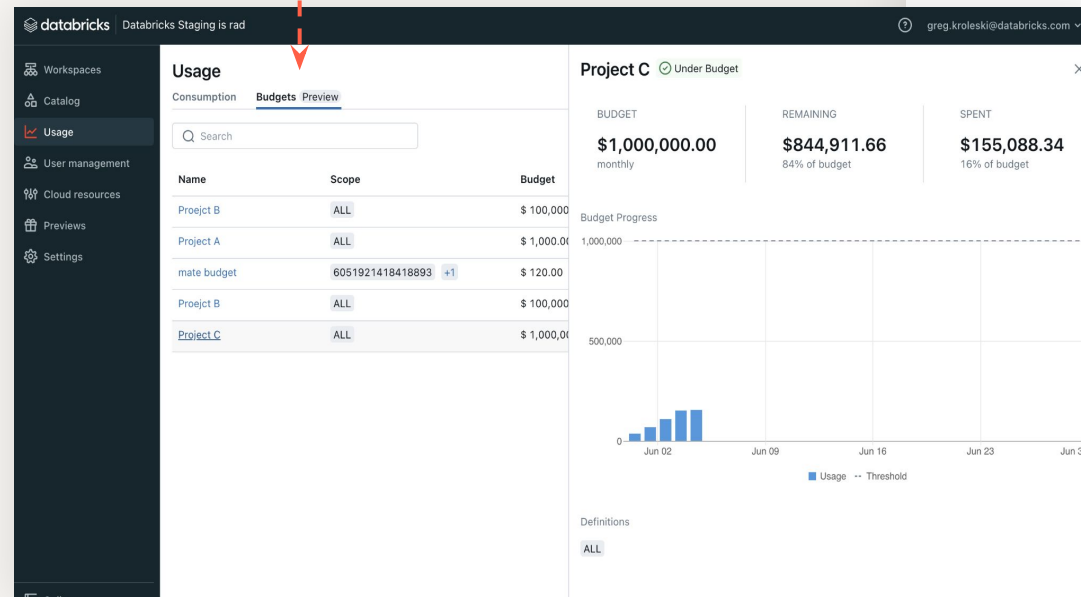
- Use 1/12 of well defined annual budget numbers
- Use last months usage + 5% (if you have history)
- Use 1/12 total annual forecast divided by the number of teams

Going forward, forecasting and budgeting will be easier with clearer historical data



How to set up budgets

- In Account Console go to Usage & click Budgets (Preview) tab
- **Create a new budget, being sure to add the relevant Tag Keys** (recommend including at least Business Unit and Project keys) Values can be left blank to include all



Usage > Budget > Edit budget

General

Name
A name for this budget.

my team budget

Amount
Specify the monthly amount in US dollars. The amount should reflect the list price for the features you use, including platform add-ons.

\$ 100000

Definitions
Define the scope of the budget. Usage must match all provided workspaces and tag filters.

Workspaces
Workspaces to be included in this budget. Leave this field empty to apply the budget to all current and future workspaces.

billing-workspace-mt-dev-aws-us-west-2-v2 X

Tags
Add tags if you want to limit the budget to usage that includes specific custom tags. Tags are case-sensitive and should be entered exactly as they appear in your usage data. To include multiple values for a key, enter values as a comma-separated list.

Key
Example: teamName

key1

key2

Values
Example: finance, dataScience, sales

abcd

Email addresses
A comma-separated list of emails that receive a notification when cumulative spending for this budget exceeds the amount in a given month. There could be up to a 24-hour delay between usage occurring and a notification being sent.

The email notification includes the budget name and tag filters. Keep this in mind when naming the budget, selecting tags, and providing email addresses.

foo@databricks.com

Cancel Create

[Detailed guide](#)

Step 3: Platform admin implementation





Applying Tags

Users should apply tags to the resources they create – Policies will enforce this

Clusters

Compute > New compute > UI preview [Send feedback](#)

New Cluster [✎](#)

Tags ⓘ

Business Unit

Add tags

Key Value

▼ Automatically added tags

Vendor	Databricks
Creator	bradford.chang@databricks.com
ClusterName	New Cluster
ClusterId	<Generated after creation>

► Advanced options

SQL Warehouses

SQL Warehouses >

UC Metric Tree Dashboard •

Overview [Connection details](#)

Status ● Running

Name UC Metric Tree Dashboard (ID: 42ae148a1f2a1839)

Type Serverless

Cluster size 4X-Large

Auto stop After 30 minutes of inactivity

Scaling Cluster count: Active 2 Min 1 Max 8

Channel Current (v 2024.40)

Tags

Budget	opex.eng.data
Service	Metric Tree

Created by [Amir Issaei](#)

Jobs

[Runs](#) [Tasks](#)

Start date [📅](#) [< Previous](#) [Next >](#)

Runs

Run total duration

2h 1m 52s

1h 0m 56s

Nov 17 Dec 29

refresh-query

Tasks

Job details

Job ID 701670430851907 [🔗](#)

Creator [Stephanie Rivera](#)

Run as ⓘ [Stephanie Rivera](#) [✎](#)

Tags ⓘ

Mid-Central MFG [✎](#)

Description

[Add description](#)

Enforcement Options

Cluster Policies

Classic Jobs
Classic All Purpose

Warehouses

DBSQL
DBSQL Serverless

Serverless Policies

Serverless Interactive
Serverless Jobs
Vector Search
Model Serving



Cluster Policies

Today users and admins enter keys and values manually as **free-text**. A set of policies should be in place to set **fixed keys and values** that map directly to your tagging policy. Admins can configure ACLs that limit cluster policies to specific users and groups.

```
{"custom_tags.PROJECT":  
  {"type":"allowlist",  
   "values":["Armadillo", "BlueBird", "Rhino" ]  
  }  
}
```

Cluster Policies

Admins can configure these policies using the UI or the API. Cluster policies are also great for enforcing cluster creation best practices. For example, requiring auto-termination.

Clusters / Cluster Policies / Create Policy

Create Cluster Policy

Cancel Create

Name

medium sized cluster

Definition Permissions

```

1 {
2   "instance_pool_id": {
3     "type": "forbidden",
4     "hidden": true
5   },
6   "spark_conf.spark.databricks.cluster.profile": {
7     "type": "forbidden",
8     "hidden": true
9   },
10  "autoscale.min_workers": {
11    "type": "fixed",
12    "value": 1,
13    "hidden": true

```

Create Cluster

New Cluster

Cancel Create Cluster

1-10 Workers: 30.5-305.0 GB Memory, 4
1 Driver: 30.5 GB Memory, 4 Cores, 1 DB

Policy ?

medium sized policy

Cluster Name

Please enter a cluster name

☐ Use your own Docker container ?

Advanced Options

DEMO

Warehouses

Warehouses enforcement is done by limiting creation to a set of user, service principals, or groups. Upon creation, those individuals ensure proper setup including; keys–value pair tags, statement timeouts and sizing.

New SQL warehouse

×

Name

SQL warehouse name

Cluster size ⓘ

X-Large

80 DBU / h

▼

Auto stop

☒ After

10

minutes of inactivity.

Scaling ⓘ

Min.

1

Max.

1

clusters (80 DBU)

Type

☒ Serverless ⓘ
 ☐ Pro ⓘ
 ☐ Classic

ⓘ Serverless SQL warehouses contain all advanced features and are Databricks' fastest warehouse type.

Prices are reduced (up to 20%) until Jan 31, 2024. Try a Serverless SQL warehouse today!

[Learn more](#)

Advanced options >

Cancel

Create

Serverless Budget Policies

Admins can configure these policies using the UI. They are then assigned to users and automatically applied to the user's newly created resources; serverless notebook, job, pipeline, or serving endpoint

Workspace settings > Compute > Serverless budget policies >

Create budget policy

[Learn more](#)

General

Name

Workspaces

The workspaces that this budget policy applies to. If empty, the policy applies to all workspaces.

Select workspaces

Tags

Used for attribution. Serverless budget policies only support fixed tags.

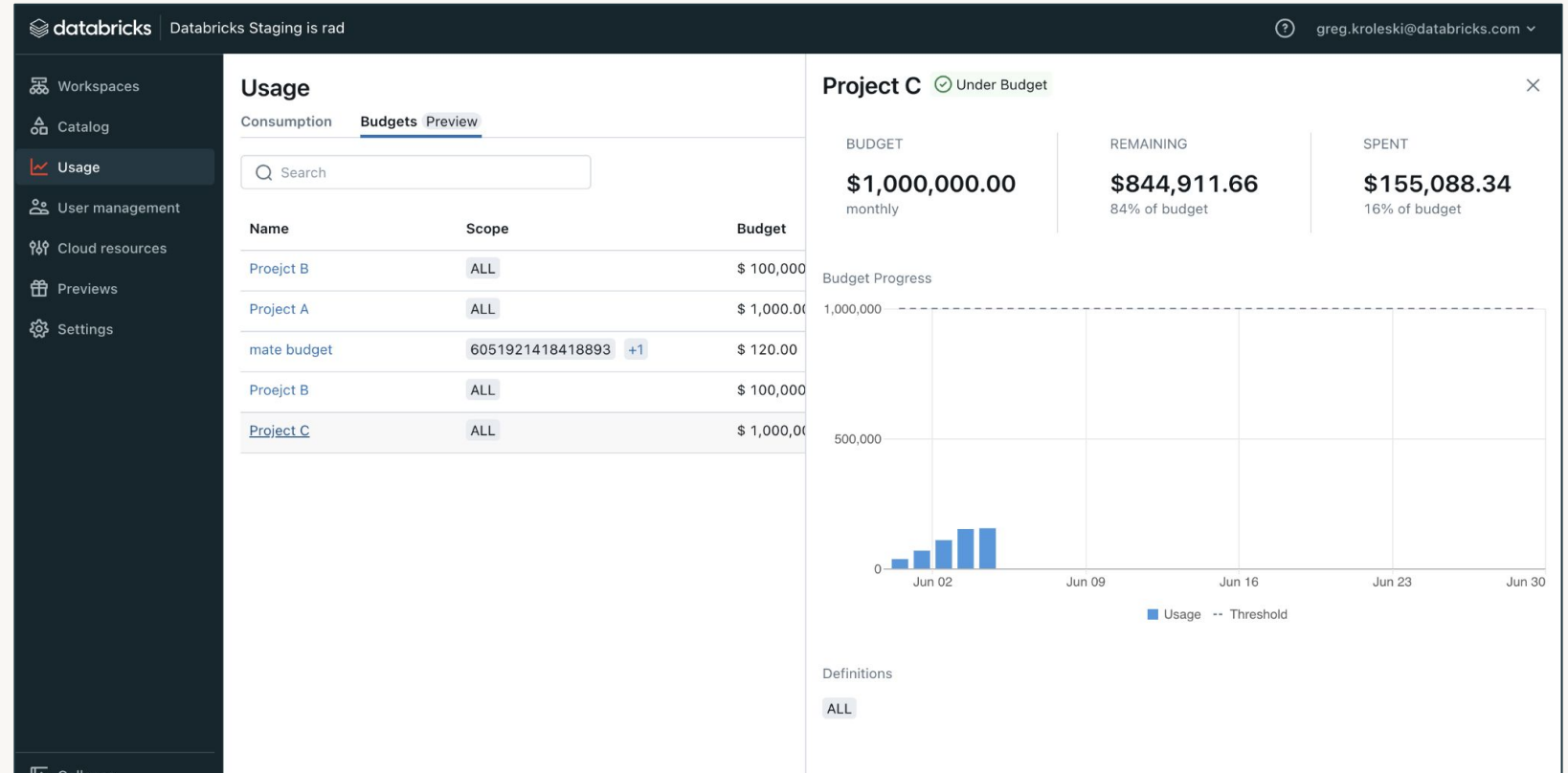
Tag	Value
No tags	
+ Add tag	

Create

Configure budget tooling

As discussed previously

Admin will create budgets in the account console as previously determined by leadership



Creating a budget

1. Budget name
2. Budget amount (\$12,000)
3. Definition
 - a. Workspaces (optional can leave blank)
 - b. Tags (key + value)
4. Email to alert

List budget amount =
Budgeted amount / (1-discount)

Usage > Budget > Edit budget

General

Name
A name for this budget.

Team Crimson

Amount
Specify the monthly amount in US dollars. The amount should reflect the list price for the features you use, including platform add-ons.

10% → \$13,333

Definitions

Define the scope of the budget. Usage must match all provided workspaces and tag filters.

Workspaces
Workspaces to be included in this budget. Leave this field empty to apply the budget to all current and future workspaces.

billing-workspace-mt-dev-aws-us-west-2-v2 × Can leave blank

Tags
Add tags if you want to limit the budget to usage that includes specific custom tags. Tags are case-sensitive and should be entered exactly as they appear in your usage data. To include multiple values for a key, enter values as a comma-separated list.

Key	Values
Example: teamName	Example: finance, dataScience, sales
Team	(Can leave blank to include all)
key2	abcd

Alerts

Email addresses
A comma-separated list of emails that receive a notification when cumulative spending for this budget exceeds the amount in a given month. There could be up to a 24-hour delay between usage occurring and a notification being sent.

The email notification includes the budget name and tag filters. Keep this in mind when naming the budget, selecting tags, and providing email addresses.

team_crimson@company.com

Cancel Create

Step 4: Example communication



Sample Communication

Account Admin Should inform their teams/users of Tagging Expectations

Subject: **[Action Required]** New Databricks tagging policies

We have implemented a new tagging practice which will be required on all new resources (clusters/jobs/sql warehouse/serverless sql) created. You will be prompted to enter tag values at the time of creation. For current running resources, teams have been given until *date* to add the minimum required tags (shown below).

Key	Business Unit
Value	101 (finance)
	102 (legal)
	103 (product)

Key	Project
Value	Armadillo
	BlueBird
	Rhino

Other encouraged tags are: Environment (dev/test/prod), Product, Milestone. Employees with permission to see system tables will be able to view the usage broken out by tags.

Please see the documentation for details. If you have any questions, reach out to me or our team at Databricks: [TechnicalPOC\(SA\)@databricks.com](mailto:TechnicalPOC(SA)@databricks.com), AccountExecutive@databricks.com



Subject: [Action Required] New Databricks tagging policies

We have implemented a new tagging practice which will be required on all new resources (clusters/jobs/sql warehouse/serverless sql) created. You will be prompted to enter tag values at the time of creation. For current running resources, teams have been given until *date* to add the minimum required tags (shown below).

Key	Business Unit
Value	101 (finance)
	102 (legal)
	103 (product)

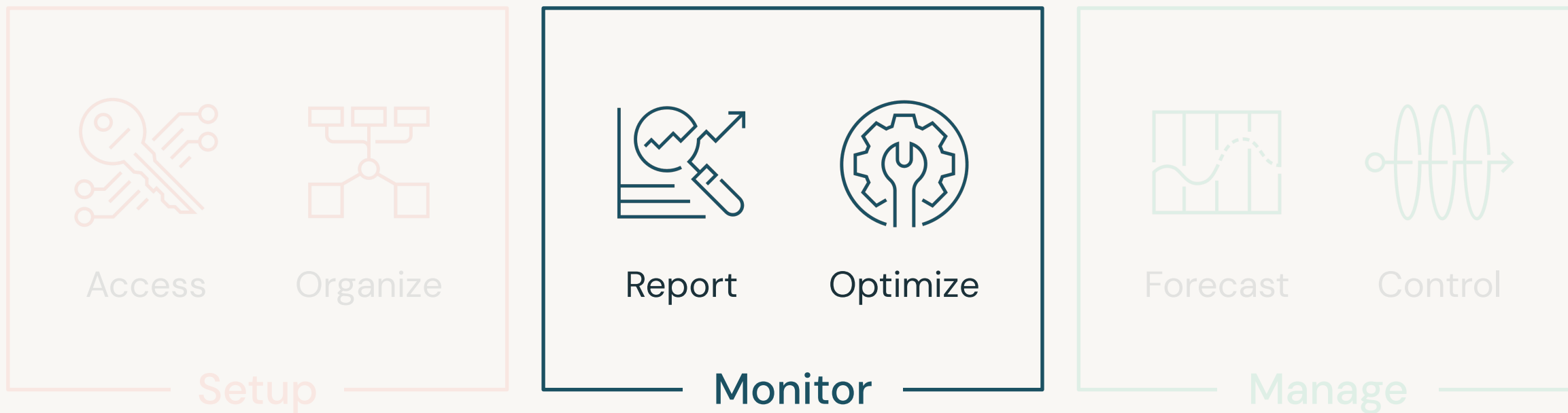
Key	Team
Value	Violet
	Indigo
	SI-Accenture

Key	Project
Value	Armadillo
	BlueBird
	Rhino

Other encouraged tags are: Environment (dev/test/prod), Product, Milestone. Employees with permission to see system tables will be able to view the usage broken out by tags.

Please see the documentation for details. If you have any questions, reach out to our team at Databricks: [TechnicalPOC\(SA\)@databricks.com](mailto:TechnicalPOC(SA)@databricks.com), AccountExecutive@databricks.com

Begin monitoring





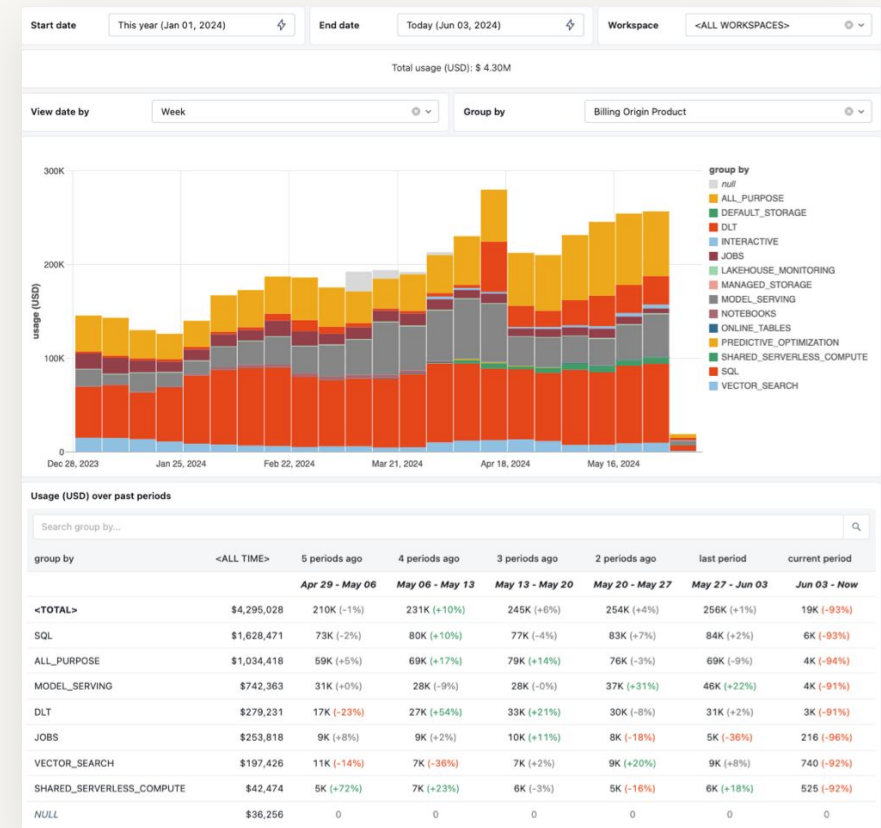
Monitoring costs & tagging (report)

Steps to set-up cost and usage dashboard in account console

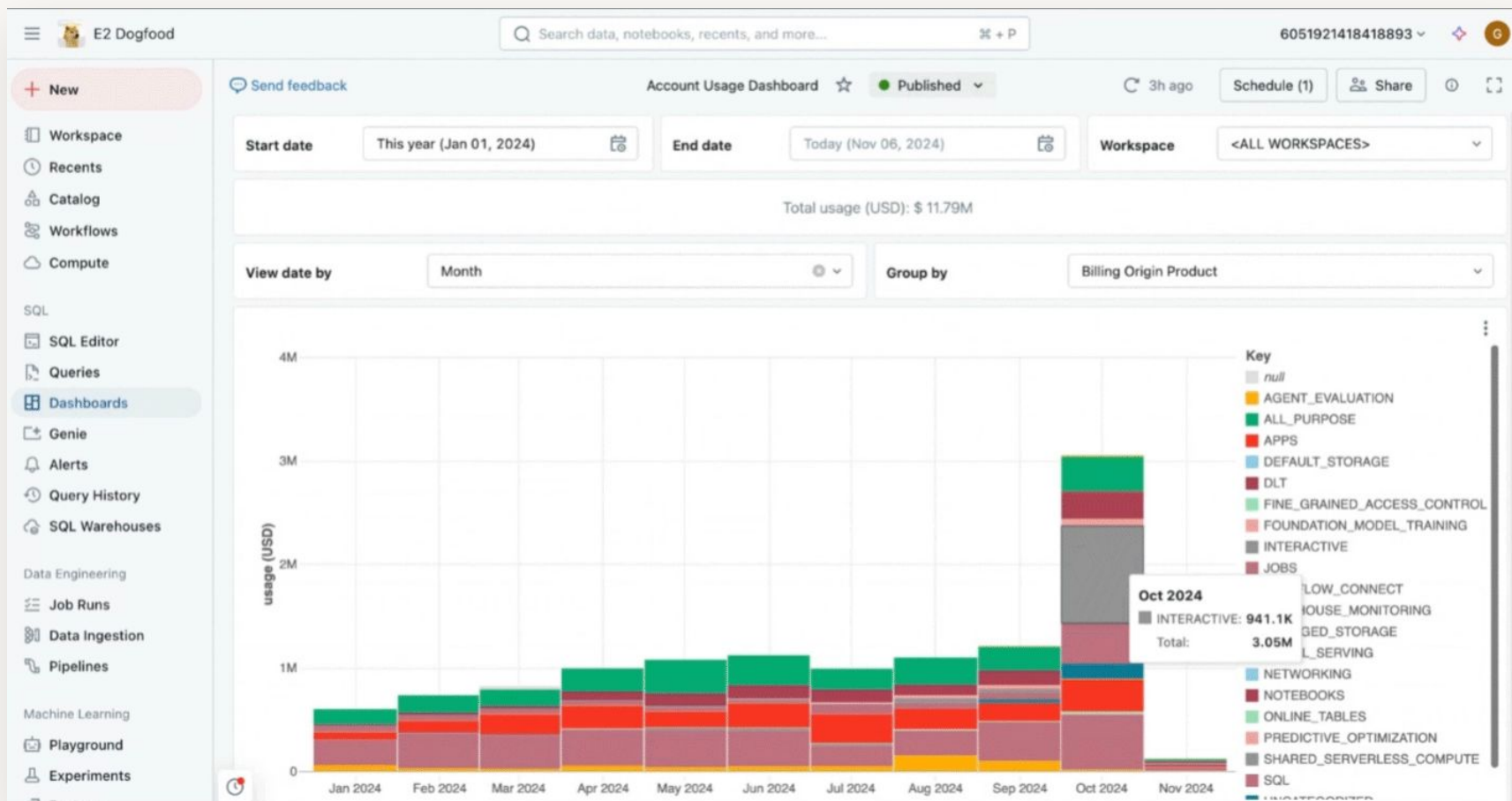
Usage monitoring in account console:

- Import cost management dashboard into your Unity Catalog-enabled workspace
- Explore usage by workspace, SKU, tags

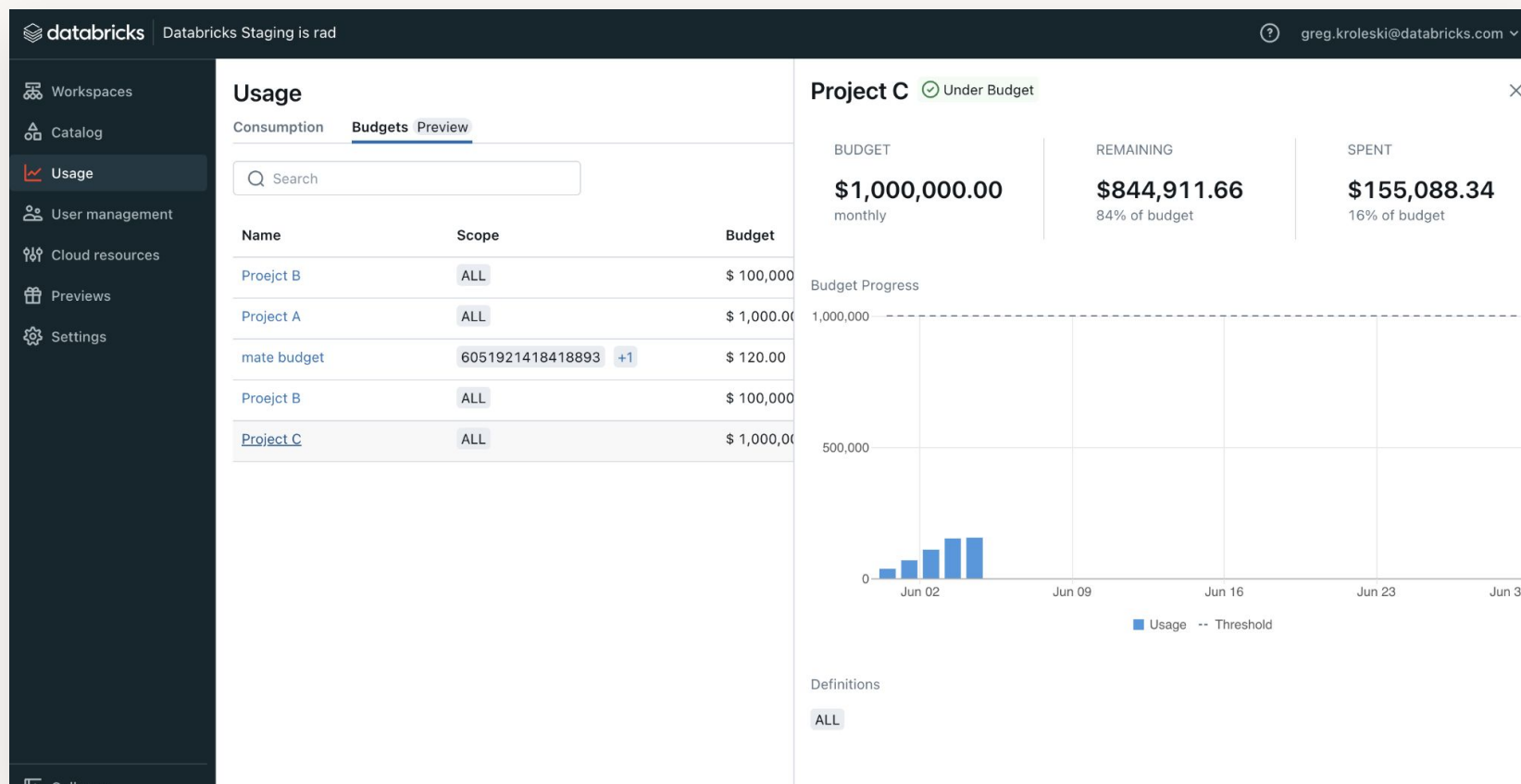
[Detailed guide](#)



Cost and usage dashboard in account console



Budget monitoring/reporting

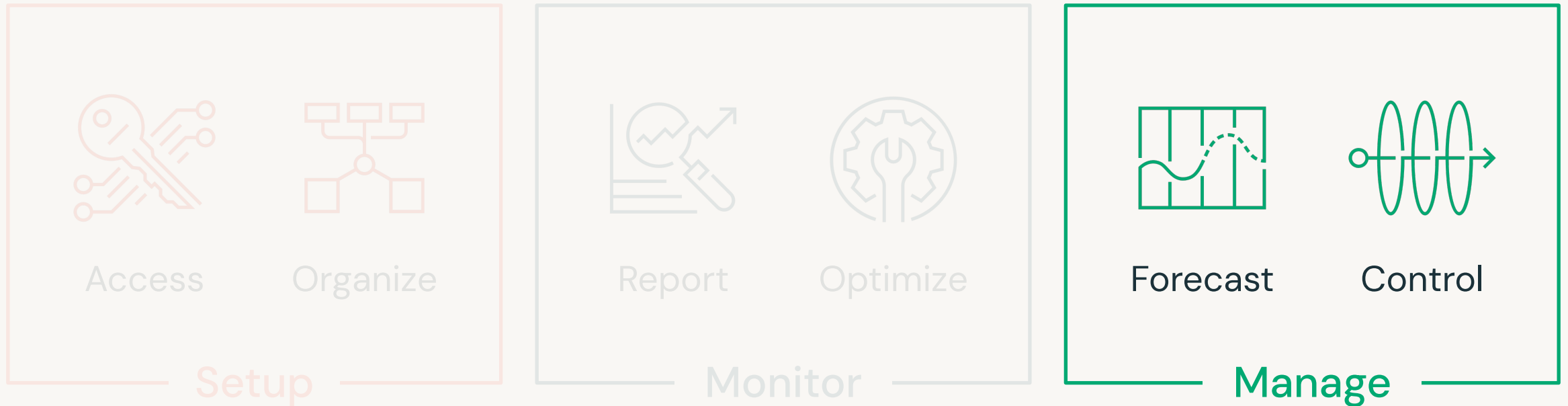


Detailed guide

As an Admin, budgets can be reviewed in account console and shared via alerts function on Budgets



Forecasting , housekeeping, iterate





Develop a housekeeping job(s)

Automating the cleanup of tags and notifications of compliance failures

Build a scheduled Job that considers the following:

- Incremental Updates: only adjusts non-compliant resources / tags vs all tags
- Error Handling: has a robust error handling to ensure the job continues even if issues occur with specific resources.
- Compliance: Ensure your tag management aligns with your organization's tagging policies and compliance requirement
- Performance: Use efficient querying and batching when interacting with large numbers of resources to optimize job runtime
- Audit Trail: Maintain a log of tag changes for auditing purposes



Managing (Cost Control)

Areas to consider

1. Use Cluster Policies to control which users are able to create clusters, as well as the size and scope of those clusters
2. Design your Lakehouse to minimize infrastructure costs, such as storage and networking costs
3. Use monitoring tools in order to make sure your expectations of cost are being met and that you have effective practices in place

Review Cost Management Best Practices



Additional resources

Where to find additional information & get support on cost management best practices

Databricks docs pages

- Setup tagging best practices and rules using Business Unit and Project as the two foundational tag keys
 - (Guidance docs: [AWS](#) | [Azure](#) | [GCP](#))
- Consider implementing fixed policies requiring only pre-defined tags to be applied to workloads created
 - (Guidance docs: [AWS](#) | [Azure](#) | [GCP](#))
- Develop budget alerts to monitor usage associated with the appropriate tags
 - (Guidance docs: [AWS](#) | [Azure](#) | [GCP](#))
- Review usage in the cost management dashboard
 - (Guidance docs: [AWS](#) | [Azure](#) | [GCP](#))

Support resources

Interested in expert guidance to implement Cost Management best practices?

- Contact your account team for available service offerings

Optimizing Costs & Usage

Identify the areas of highest spend for further review

Overview of DB Cost Optimization

- Confirm the right resources are being used for the specific needs (guide)
- Consider autoscaling / auto-termination / format optimization (guide)
- Prioritize areas for optimization and work with development teams to drive optimization changes



Setup Unity Catalog and metastore

Check if Unity Catalog is setup & if metastore is active:

Account admin:

- Log into the account console
- Click "Workspaces"
- Find your workspace and check the "Metastore" column
- If a metastore name is present, your workspace is attached to a Unity Catalog metastore

If not:

- In account console go to data section and select 'Create Metastore' pick your desired workspace and activate unity catalog

[Detailed guide](#) to enable system tables schema

Budget alerts setup



Databricks Staging is rad

Usage

Consumption **Budgets** Preview Consumption (Legacy)

Search Add budget

Name	Scope	Budget	Progress	Spent	Remaining	Status
Greg's Budget	dais-cleanroom-demo1	\$	-	-	-	No data
Personal Budget	volumes-dais24	\$	-	-	-	No data
Streaming UK	workspace +1	\$	-	-	-	No data
Project A	Lighthouse Test LHA	\$		\$	\$	Under budget
2024 Commitment Spend Plan	ALL	\$		\$	\$	Under budget
Mate's Budget	6051921418418893 +1	\$	-	-	-	No data
Project B	Data Marketplace Provider DAIS	\$		\$	\$	Exhausted

